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# Impact of Open-Source Community on Cryptocurrency Market Price: An Empirical Investigation

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**ABSTRACT**
Although the prices of cryptocurrencies remained volatile for the past decade, the factors that impact the price dynamics of the new type of investment instrument have not been fully identified yet. In this study, we recognize the dual nature of cryptocurrencies, that is, being a software program and a financial instrument, and examine the impact of software advancement on the price dynamics of cryptocurrencies. The open-source software (OSS) platform functionality enables social behaviors that we use as signals. Using data from the largest OSS platform, we establish the connection between open-source activities and the price movement of cryptocurrency. In particular, as project popularity (forks and watches) and users' feedback (issues) increase, the market price increases by 4.3 percent, 2.4 percent, and 4.4 percent per annum, respectively. On the contrary, the number of code corrections (pull requests) is negatively related to prices leading to a 5 percent annual price decrease. Our results suggest that OSS contributions create a perfect selection mechanism, where higher quality projects receive more developers' attention and user feedback, whereas lower quality projects do not, thus creating the separating equilibrium.

**KEYWORDS**
Cryptocurrency; open source software; blockchain; signaling theory; OSS; online platforms

## Introduction

Since their emergence in 2008, cryptocurrencies have become a part of the modern financial landscape. The adoption of Bitcoin, the first publicly known cryptocurrency, has reached 82.2 million wallets as of 2022 [129]. Moreover, Coinbase, one of the biggest U.S. marketplaces to trade cryptocurrencies, serves 108 million verified investors worldwide [35]. In the wake of crypto adversities, investors' and authorities' attention has turned to seek investment risk reduction policies, that is, information disclosure, due diligence, and reporting, for the cryptocurrency segment of the financial market.

Several major exchanges (e.g., FTX), crypto lending platforms (e.g., Voyager, Celcius, and BlockFi), and hedge funds (e.g., Three Arrows Capital), have experienced severe economic damages that led to their bankruptcies and losses of clients' funds [43, 97]. Total debt accrued by the crypto industry in 2022 is evaluated at over \$18 billion to over 1 million creditors.[1] While the latter companies have mostly suffered from the liquidity crisis and mismanagement of financial assets, the collapse of the LUNA token in early 2022

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has been provoked by the emission algorithm deficiency that several wallet addresses have exploited [112, 120]. Those instances of failures have increased the need for a new regulatory approach to crypto assets. In the recently introduced Responsible Financial Innovation Act [120], U.S. senators refer to GitHub users as major stakeholders in the crypto industry and invite them for their input on future crypto regulation. Moreover, recent research raises the question of the fiduciary role of digital currency developers [136], which emphasizes the need for greater attention to the software development process.

Both academic and practitioner communities have long pointed to the importance of software code in cryptocurrency valuation. First, the software nature of cryptocurrencies has been highlighted in popular media and academic research [66, 93, 95]. Open-source software (OSS) metrics are reported on major websites, such as CoinGecko,[2] CoinMarketCap,[3] and popular social media (Online Supplmental Figure 1.1). These resources facilitate knowledge transfer and help resolve uncertainty in investment decisions. Therefore, our study aims to create a theoretical background for the impact of OSS development as a cryptocurrency fundamental and provide empirical evidence of the direction of the impact of the OSS metrics.

In this paper, we point our attention to the software nature of cryptocurrencies and their programmed code as a novel signal of the fundamental quality of digital financial assets. We treat cryptocurrencies as software projects, on the one hand, and financial instruments, on the other hand. As a financial instrument [39], cryptocurrencies are publicly traded on the global markets and have the market infrastructure that supports crypto investment activity, including derivative products, such as Bitcoin ETF, crypto swaps, and DeFi. From a software perspective, cryptocurrencies rely on complex algorithms to reach a consensus on the state of the transactions that are programmed in code and involve extensive computations [1]. The novelty and complexity of technology may affect the dynamics of other factors that constitute cryptocurrency quality, including its security. We devote the Online Supplemental Appendix 3 to discuss cryptographic algorithms.

The unique feature of blockchain-based currencies is that 66 percent of projects are OSS and have publicly available information about their software development on major OSS platforms, like GitHub.[4] The importance of the OSS activity finds support from the practitioner investors community as an important signal for the due diligence process [25, 90, 107]. Moreover, Rohr and Wright [116, p. 463] suggest that failure to list code in an open-source site "may signal ulterior motives on the part of the party selling the token."

Prior academic literature has discussed the importance of the OSS activity for entrepreneurial success [58, 61, 79, 109, 133, 135], however, not in the financial success context. The cryptocurrency context uniquely allows us to explore the connection between OSS activity and the financial performance of the software project. Investment decision-making has been linked to the learning processes that resolve uncertainty about the asset [50]. Wider literature on OSS development emphasizes the use of software information for the assessment of project legitimacy and risk [91, 124]. Recognition of the OSS platform and its users is an important element of the cryptocurrency infrastructure, and therefore, activity on the source code is emphasized to be valuable information for potential investors.

We use signaling as an overarching theory [127] to explain the mechanism of the role of software development efforts on cryptocurrency market performance. Projects with inherently greater potential attract development talent that contributes to development activity for the project's growth and quality [86]. OSS activity is connected to the development efforts that are associated with software quality improvement [64,77]. By observing the quality improvement effort, investors resolve the existing uncertainty and update their beliefs about the quality of the software and cryptocurrency fundamentals. The investor sentiment about digital asset fundamentals affects the demand and leads to market price changes [106, 123]. In addition, the competition between software projects for the limited development talent is led by the project's perceived quality [49], which makes the signaling mechanism more credible. Therefore, by observing the software development effort, consumers can make an inference about the quality of the digital product, or cryptocurrency. Eventually, consumers would prefer the higher quality product to the lower quality product [132].

Our study contributes to a growing body of literature that examines the phenomenon of cryptocurrencies and blockchain-based systems and their market behavior. We point out the dual nature of cryptocurrencies as publicly traded financial assets and software projects. Such duality is unique, and the ability to measure its market characteristics (price) as well as development activity (OSS contributions) distinguishes cryptocurrencies from other digital products. Furthermore, it connects our study to the literature focused on market signals and OSS contributions. By offering a novel source of signaling information in the digital tokens setting, we propose a novel framework for decision-making that extends to other digital products. Our results suggest that OSS contributions create a perfect selection mechanism, where higher quality projects receive more developers' attention and user feedback, whereas lower quality projects do not, thus creating the separating equilibrium. In addition, we find novel evidence that the point of separation between high-quality and low-quality projects may vary over time. The results of this study have important implications for investors and policymakers.

## Background and Theory Development

Our paper offers a novel view of cryptocurrencies as financial instruments, on the one hand, and software projects, on the other hand. Therefore, we rely on the prior cryptocurrency, signaling, and OSS literature to develop our theoretical framework for cryptocurrency quality signals.

### *Cryptocurrencies*

Unlike traditional financial assets, cryptocurrencies do not have an underlying business entity and cash flows. Therefore, the evaluation of the cryptocurrency business model remains uncertain, and the current literature has a rather limited view of the factors that impact the cryptocurrency market price [33, 36, 37, 69]. Extant research also has mainly focused on Bitcoin price [59, 98], overlooking the less dominant but numerous alternatives to Bitcoin, or altcoins. These limitations motivate us to conduct our analysis.

The early studies on the emerging market of cryptocurrencies discuss the classification of cryptocurrencies as financial instruments [13, 23, 24, 34, 45, 134] and assets [44, 54, 55, 92, 139]. From the financial instrument perspective, early cryptocurrency studies explored the impact of traditional macroeconomic factors [89] and public sentiment [22, 98] on the market performance of the novel digital assets. In addition, prior literature has discussed the challenges of high volatility [82], the questionable utility of the currency [55], and information asymmetry between investors and project teams [68].

As digital assets, cryptocurrencies greatly rely on the underlying technology. From the technological perspective, later works appealed to the technology determinants of cryptocurrency exchange rates, such as mining technology and mining difficulty [89]. Despite the prior investigation, the IS theory remains inconclusive about the technological nature of cryptocurrencies and the role of software-development factors in their market performance.

Therefore, in our paper, we explore an alternative source of fundamentals of the quality of cryptocurrency as a digital asset that contributes to filling the market with accurate and timely information and market efficiency. We point at the signals from the OSS development community as easily accessible and observable information about the crypto project, which can be quantified and used by market participants for the informed decision-making process. Eventually, the availability of high-precision information about the asset class improves the optimal allocation of resources and market efficiency.

## Signaling Theory

Signaling theory describes the decision-making process in situations of information asymmetry [127]. The basic signaling model suggests that employees know their true productive capabilities—low or high—and employers do not have this information. In our case, crypto projects are "employees" with varying technological characteristics, and investors are "employers" that pick a currency without full knowledge of its inherent qualities. Despite the uncertainty, the investor gets a sentiment about the project's private information in the form of observable characteristics, such as the level of software development. Over time, an investor will learn the real value of their investment and will update their beliefs. We suggest that the higher OSS activity is correlated with more positive investment sentiment.

Similar to the literature on financial signaling exemplified by the work of Bhattacharya [17, 18], Leland and Pyle [84], and Ross [117, 118], our paper looks at how firms signal the net value of their anticipated growth opportunities. Prior research in finance hypothesized that managers tend to signal the value of the firm through the capital structure [117], dividends [17, 75, 103], or conversion of an outstanding convertible bond [63]. Due to the absence of the corporate structure, these metrics are not available in the cryptocurrency setting, which motivates us to search for the signals inherent in the digital asset setting. Later studies demonstrate that the reputations of investment bankers [28], auditors [14], boards of directors [29], venture capitalist and angel investor presence [47], and top management team [88] serve as signals in the IPO process. Due to the decentralized organizational structure, the institutional indicators are barely applicable in a cryptocurrency signaling setting. To our best knowledge, this study is among the first to treat software development intensity as a signal for the quality and market value of digital assets. We further discuss how the OSS signals meet the observability and credibility criteria and create a separating equilibrium.

Separating equilibrium is the central concept of the signaling theory and refers to a situation in which agents with different characteristics choose different actions. Following the example in Spence's seminal paper [127], a high-quality worker will get a higher level of education (higher effort), and a lower-quality worker will choose a lower level of education (lower effort). This constitutes the separating property of the equilibrium. Due to a high-quality worker requires lesser effort to get a higher level of education than a low-quality worker and given that the higher-quality work is paid better than lower-quality, neither of the workers will change their strategies. This constitutes equilibrium. In our context, the high-quality project will have a larger development effort and the low-quality will have less. In the next paragraphs, we elaborate on important properties of the OSS activities that make them proper signals and how due to these properties they create a separating equilibrium.

First, over 66 percent of cryptocurrencies have their code publicly available on the major OSS platforms that are easily accessible by investors and software developers. Moreover, major media and analytical outlets, such as CoinGecko, CoinMarketCap, and Santiment, report the OSS activities dynamics that make them easily observable to investors. The saliency of OSS-related information in practitioners' media channels emphasizes the importance of software development metrics in the due diligence process. The lack of a public software audit led to irreversible damages to investors in the case of FTX and Alameda fraud [16]. The pervading "rule of code" in the digital age [93] sets the agenda for modern investors to understand how the software functions and is being developed.

Second, the inherent mechanisms of the pull-based development make the observed OSS activity more creditworthy than other popular information sources. Imitating OSS activity is costly, especially over a long period of time. Some of the activities are harder to imitate than others, and this is one of the reasons to take into account the scope of metrics rather than focusing on just one. On the one hand, high-quality projects would naturally attract developers and their contributions. On the other hand, from a cryptocurrency project perspective, imitation of OSS activity is costly for low-quality projects as they need to generate fake activity continuously and across multiple different activities. While some activities may be easier to generate by making nuisance changes to the code lines, like committing spaces to the files, other activities are of social nature, like forks, and would require multiple account creation that is controlled by the platform. The GitHub platform has verification policies that aim to detect "fake" accounts and protects the platform. According to GitHub's Acceptable Use Policies, "the staff may take further restrictive action against accounts that are engaging in these types of behaviors" [51].

From a cryptocurrency investor perspective, imitation of OSS activity by generating nuisance activity is limited and costly. The limitation comes from the peer-review process of the OSS development, and costliness comes from the high effort needed to circumvent the peer review. Prior studies report that trust in open-source software depends on the peer-review process [61, 122]. Peer review filters out low-quality coding, or nuisance coding, from legitimate fixes that improve the quality of the software. For example, if investors wanted to imitate commit or pull request activity, they would need to set up git,[5] which requires basic programming knowledge. In addition, to imitate the fork or watch activity, the investor would need to do it from multiple accounts that are regulated by the platform [51]. Therefore, we may conclude that imitation activity is highly unlikely for an average investor.

Third, we explain how the OSS signals create a separating equilibrium. We consider that developers registered on the platform, due to their expert knowledge and professional experience in technology, have a unique ability to recognize quality and would not invest their costly time and effort in low-quality projects [86]. Thus, they play the role of the third party that is willing to take on the costs of signaling that creates a separating equilibrium [15, 60]. According to signaling theory [127], a credible signal can create a separating equilibrium, or a market outcome in which the project with a higher OSS activity will credibly signal its quality to the market, whereas the project with a fewer OSS activity will not be able to do so.

Cryptocurrency software projects are heterogeneous in their potential to attract development talent: High-quality projects with clear technological objectives and development strategies are more interesting for developers and attract development talent [86]. Second, platform regulation and peer review processes in the open-source community make low-quality digital tokens difficult to create fake open-source activities. Therefore, developers' activity represents an efficient signal due to its observability [38] and differential costs for high- and low-quality tokens [19].

## OSS Contributions

We rely on a significant body of open-source literature [48, 87, 105, 115, 137]. Most works in these areas converge in their discovery that intrinsic factors as image, reputation, recognition, intrinsic pleasure, and willingness to help [83]. In addition, modern open-source software (OSS) platforms incorporate many social features such as following, liking, and commenting on someone's work. Such platforms propagate social behaviors that affect the contribution dynamics [108]. Moreover, developers who seek to contribute to software projects rely on multiple signals from the activity in the project community. For example, developers interpret activity traces—following, watching, and committing—as an important signal of community support and an indicator of high project quality [41].

From the software user perspective, the quality is a combination of the bugs in the product and the efforts the vendor makes to patch the bugs after release [9]. Several studies in the information systems (IS) literature establish the close ties between the software quality and the amount of product development effort [70, 78]. Consequently, the development effort can be perceived as the proxy for software quality. Due to the ongoing peer review process, open-source projects improve quality and can eventually succeed in the marketplace (e.g., Linux OS).[6]

Prior studies have also related the amount of the OSS contributions and the measures of software success [40, 73, 100, 101]. On the one hand, the IS literature documents the impact of open-source activities, such as the number of commits, pull requests, and issues, on software project success [125]. On the other hand, prior studies define the OSS project success as the number of commits [58], a project use-value [115], or the number of downloads [101], due to the lack of monetary indicators of the market success [100]. The unique context of our study allows us to measure cryptocurrency software success by market indicators. Despite their underrepresentation in the traditional OSS literature, market measures of market success are suitable. As Lerner and Tirole [87] suggest, developers' long-term incentives for working on an open-source project are stronger when their contributions may lead to future access to the funding market. In the case of cryptocurrencies, funding comes from investors who purchase crypto coins and tokens.

Given that the development effort, such as commits, pull requests, etc., is the proxy for software quality, its impact on the software's success can be described using signaling theory as follows. A high development activity implies that (1) the project is on track to fulfilling its promise to the customer to deliver functional and secure digital products (cryptocurrency), (2) the project is likely to ship new features in the future that can benefit the cryptocurrency holders, and (3) it is less likely that the project is just a scam. The OSS inputs and the financial performance are connected though the perceived quality mechanism [71, 114]. By observing the development effort, investors update their beliefs about the quality of the software and reduce uncertainty about the cryptocurrency. The investor sentiment about digital asset quality affects the demand and leads to market price changes [106, 123].

Our research tracks the five most popular activities from the OSS platform: commits, pull requests, issues, forks, and watches—to assess the development effort. These metrics are the focus of the software developers [41] and are frequently used to assess the quality of the project on the OSS platform [40, 73, 100, 101, 125]. On the one hand, we may expect that code contribution activity has a positive effect on the cryptocurrency's market price. First, projects with high development activity would attract more developers [42], improving source code quality and increasing investors' expectations about the product [126]. Second, investors would adjust their sentiment according to their observation of high innovation activity and community engagement, and stock prices start moving [106, 123]. On the other hand, a high level of software activity may be generated by code corrections or debugging due to the number of pitfalls detected. Consequently, the direction of influence is not apparent and needs further investigation. To provide insight and narrow the gap in the extant literature, we formulate our main research question:

***RQ:*** *How do different types of development activities in an open-source community influence the price dynamics of cryptocurrencies?*

To address our research question, we examine the five most popular activities from the OSS platform: commits, pull requests, issues, forks, and watch events. These metrics are often used to assess the quality of the project among the developers [41] and are discussed later in detail. We construct monthly aggregated variables of the OSS metrics to identify the significance of each type of activity and their effect on the market price. Monthly aggregates are the most used to model prices [27, 74] since they absorb excessive volatility and yet allow to capture the changes in the market prices. In addition, we control for cryptocurrency-specific factors using a fixed-effects model. We also use the instrumental variable (IV) approach to address the concern of reverse causality and endogeneity of the OSS development metrics. Our estimation results are robust.

## Data Description

### *Data Setting*

For our analysis, we collect data about cryptocurrencies' OSS accounts from one of the biggest listings of cryptocurrencies: CoinMarketCap[7] (Online Supplemental Figure 1.3). Links to OSS profiles of cryptocurrencies are widely observable and available on other sources, for example, Coin360 (Online Supplemental Figure 1.4) and CoinGecko (Online Supplemental Figure 1.5). We use the data from GitHub, which is one of the most popular platforms with 28 million accounts and 85 million repositories,[8] and over 66 percent of cryptocurrency projects host their code on GitHub. GitHub is an online repository hosting service that allows collaboration on software development projects. GitHub also offers a version control system, which allows contributors to make changes to the same files concurrently and enables them to revert a document to a previous revision, track each other's edits, correct mistakes, etc. GitHub allows businesses and open-source projects to collaborate through shared accounts called organizations (see the Online Supplemental Figure 1.6).

To contribute to projects on GitHub, one should complete registration and have a personal account. Once the account is created, GitHub users may contribute to the project on GitHub. Each platform user has a personal profile and can own multiple repositories, public or private. A repository is the folder of the project (see Figure 1). A repository contains all the project files and stores each file's revision history. Repositories can have multiple collaborators and contributors and be either public or private. A major share of cryptocurrency accounts is public, allowing easy access for platform users willing to contribute or explore the development progress. GitHub users can also create or join organizations and collaborate on another user's repository.

The OSS platform has enabled a widely adopted pull-based development model. Pull-based development (PBD) is a distributed development model which allows collaboration in software development [57]. The PBD model facilitates social contribution, easy testing, and peer review. In this model, potential contributors fork, or clone, the code and make changes in their local copies independently from each other. After a set of changes is made, they create a pull request and submit changes for revision by a member who has the authority to decide about the quality of the submission and their further integration into the project's main development line. With the functionality described above, a software project is designed to have a master branch that is always deployable and other branches, which are essentially copies of the master branch created by users for editing code. Since GitHub employs a peer-review collaboration model, other branches stay unmerged with the master branch until reviewed by other project members. The entire workflow chart is given in Online Supplementary Figure 1.12.

Based on the version control system design in GitHub, pull requests are associated with the branches of the repository's parent branch. One can merge proposed changes into the parent branch from another branch using a pull request [52]. For example, pull request #14171 in the Ethereum GitHub project is created from the branch "ast-import-via-standard-json" within the "ethereum/solidity" repository. If the repository

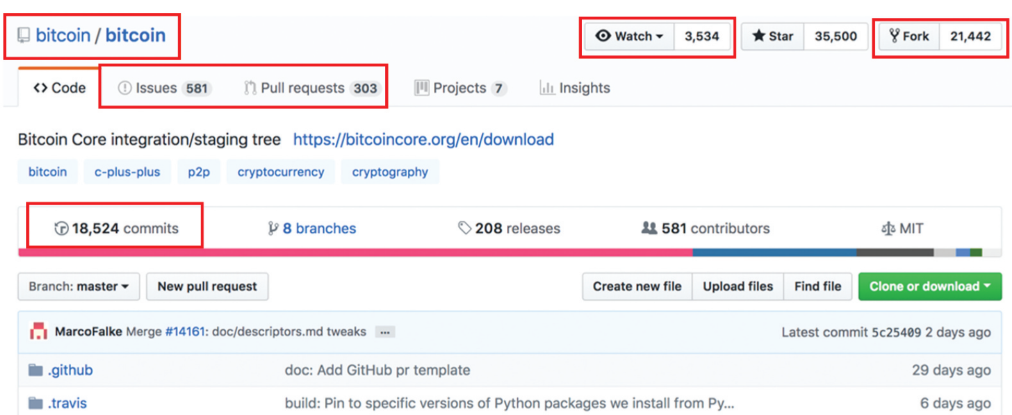


Figure 1. Repository view on GitHub.

is organization-owned, one must be a member of the organization that owns the repository to open a pull request. Outside developers, on the other hand, should create a fork, that creates a branch to do a pull request [53]. For example, pull request #688 in the 0xProject GitHub project is created from the fork. We account for this dependency in the PVAR analysis. The PVAR analysis accounts for the dynamic structural dependencies of multiple variables [3], including the relationship between pull requests and lagged variables of forks and vice versa.

The GitHub workflow process allows multiple actions to be done with the code, which is stored in repositories. Key actions that developers use while working on code are: adding commits, creating pull requests, reviewing the code, merging changes, and debugging the code. Our data contains the indicators of activity on the open repositories, such as the number of commits, pull requests, issues, forks, and watch events.

- *Commit* is an individual change to a file (see the Online Supplemental Figure 1.7). Each commit has a unique id that allows keeping track of what changes were made, when, and by whom. In a version control system like GitHub, each user can make a local copy of the repository and work it from her private computer. To make the change happen and enable others to access them, one should send or push committed changes to a remote repository hosted on GitHub.
- *Pull request* is a proposed change to a file submitted by a user and accepted or rejected by a repository's collaborators (see the Online Supplemental Figure 1.8). Pull refers to when people are fetching in changes and merging them. When a contributor wants to create a pull request, she creates a branch, a parallel repository version. It is contained within the repository but does not affect the primary or master branch. Creating branches allows working freely without disrupting the main version of the document. A pull request is created when changes are made and ready for revision. Approved and merged pull request becomes a commit.
- *Issue* is a suggested improvement, task, or question related to the repository (see the Online Supplemental Figure 1.9). Issues for public repositories can be created by anyone and are moderated by repository collaborators. Each issue contains its discussion forum and can be labeled and assigned to a user.
- *Fork* is a personal copy of another user's repository, which is attached to the account of one who forks (see the Online Supplemental Figure 1.10). Forks allow people to make changes to a copy of the project without affecting the original. After the fork is made, it remains attached to the original repository. This allows the developer to submit all changes made on the fork to the original repository or fetch updates from it.[9]
- *Watching* is a way of subscription to receive email or web notifications on a repository (see the Online Supplemental Figure 1.11). Once the user starts watching a repository, she will receive notifications for all discussions, like issues, pull requests, and commits.

We provide a mapping of each activity to its corresponding signaling property. Table 1 summarizes the OSS signals. Commits best describe the number of changes to the software code. Each commit follows a decision process of the core developers (more committed ones) to accept or reject the suggested edit of the software code [56]. A large number of commits signifies the high involvement of the decision-makers in software quality enhancement and faster product improvement [99].

**Table 1.** Signaling properties of OSS activities.

| Activity | Effect on Software | Definition | Type of Signal | Signaling Property |
|---|---|---|---|---|
| Commit | Quality enhancement | Accepted change to a file. | Quality | Intensity of software contribution |
| Pull request | | Proposed change to a file that waits for revision. | | Intensity of community involvement |
| Issue | | Suggestion or feedback on a topic. | | Intensity of user adoption and feedback |
| Fork | Diffusion of software | Personal copy of another user's repository. | Popularity | Intensity of community recognition |
| Watch | | Subscription, similar to "*follow*" on social media platforms. | | Intensity of public attention and awareness |

A higher volume of enhancing activity also helps legitimize the project in the eyes of the users and creates more trust [41, 121]. The pull requests are the proxy of community involvement and indicate the collaboration intensity of the developers willing to contribute to the project. Developers are cost-effective [10, 67, 87] and would only choose to contribute to projects of high potential. Therefore, pull requests signal the quality enhancement intensity. Issues are requests for improvement, reflecting user engagement with the project. Users of the digital product interact with software, assess it, and communicate their feedback to the developers. Software products with many issues benefit from elevated user activity since more social capital enables developers to understand user feedback and speed up the development process [58, 62]. Therefore, we suggest that these activities are considered *signals of quality*, as they affect the code directly.

Forks are a powerful indicator of the software's reputation in the community [61]. People fork the highly recognized repository when they (i) want to make a creative change to the project, (ii) want to save a copy for personal use, or (iii) want to use some or all of the code in that repository as a starting point for their project. Therefore, the higher number of forks signals greater value and recognition of the project in the community and leads to higher adoption. Watch events indicate the scope of public awareness of the project. Watching lets software users subscribe to notifications for activity in a repository and receive updates about developers' actions. Watching a project helps the platform user track the developer's activity within the project regularly and incorporate new information about the quality evolution. The higher the number of subscriptions (watches), the greater user awareness about the project. As the prior study indicated, "people are watching because they depend on it" [41, p. 1281]. Therefore, we suggest that these activities are considered *signals of popularity*, as they illustrate the amount of public attention to the project.

### Summary of Data

We collect market data from CoinMarketCap using Python Selenium and the development activities from the public GitHub Archive dataset accessible via Google BigQuery service.[10] Our dataset includes 559 cryptocurrencies active on August 31, 2018, and that have their code publicly available on GitHub. For analysis, we include cryptocurrencies that have at least four months of price observations at the moment of the data collection start.

We collect development activities for the cryptocurrencies in the sample and aggregate them in monthly observations during the period from August 2016 to December 2019. Due to the changing volatility in market data during the investigated timeframe (see the Online Supplmental Figure 1.2) and development data, we construct monthly aggregates that reduce noise, similar to Mittal and Goel [104]. The dynamic of the OSS activities for the investigated timeframe is provided in Figure 2. We use the closing price[12] at the end of the month (EOM) as our dependent variable and construct a monthly aggregated number of open-source activities for the calendar month. It is important to note that such construction ensures that OSS activities predate the time of the price change. The main independent variables in our analysis are activities widely used in the OSS literature: commits, pull requests, issues, forks, and watch events [58, 73, 101, 125]. A large variation in the scale representation of the variables motivates us to use natural logarithms. With the total number of price observations is 15,716, we describe summary statistics in Table 2.

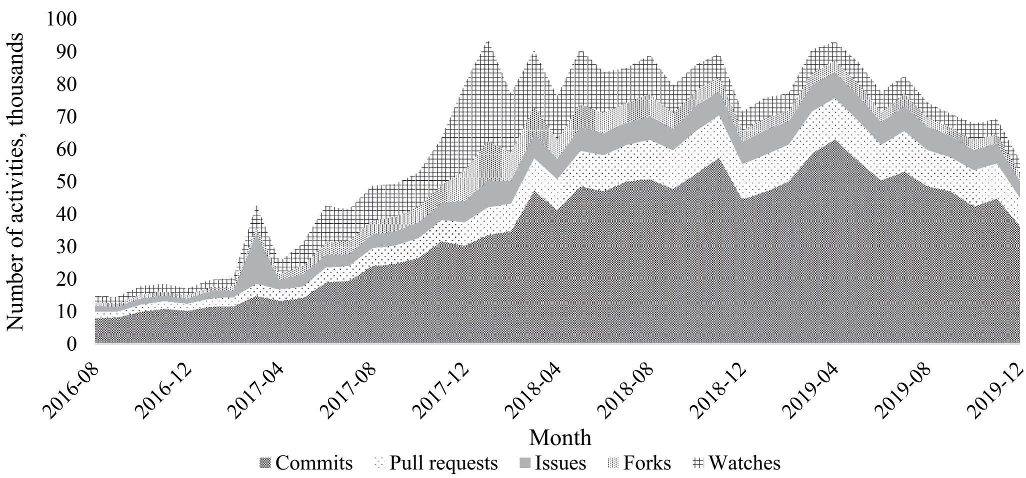


**Figure 2.** Dynamic of open-source activity across events.[11]

**Table 2.** Summary statistics of key independent variables.

| Variable | Description | N | mean | sd | min | max | Average per day |
|---|---|---|---|---|---|---|---|
| Price (DV) | EOM price, USD | 15,716 | 64.52 | 12,272.39 | 1.56e-08 | 82,624.1 | n/a |
| commits | Number of commits | 15,716 | 74.24 | 297.21 | 0 | 11,957 | 2.47 |
| pullreq | Number of pull requests | 15,716 | 18.44 | 62.54 | 0 | 1,053 | 0.61 |
| issues | Number of issues | 15,716 | 13.32 | 119.10 | 0 | 13,394 | 0.44 |
| forks | Number of forks | 15,716 | 9.04 | 61.48 | 0 | 2,220 | 0.30 |
| watches | Number of watch events | 15,716 | 18.62 | 168.60 | 0 | 7,923 | 0.62 |

Abbreviations: DV, dependent variable; EOM, end of month; USD, United States Dollar.

As we can see from the summary statistics presented in Table 1, cryptocurrency projects attract attention and development talent. On average, developers commit to each cryptocurrency project 74.24 times per month, or 2.47 commits per day. Regular commit activity demonstrates the continuous effort of the community to develop the project and improve its quality. The pull request activity is 18.44 per month, or one pull request every two days. Regular pull request activity may signify an active interest from the OSS community who are not directly related to the project but chose to contribute to the project. The average monthly number of issues is 13.32, or 1 issue every three days. The main contributors to issues are (1) identified bugs and (2) requests for new features. Both are markers of end-user demand, and a large number of issues is a good signal of the increasing popularity of the software product with the users. The average number of forks equals 9.04 per month, or one fork every four days. Such forking activity is relatively regular, and, given the high standard deviation, for some projects, it is higher than average. The average number of new subscriptions, or watch events, is 18.62, or 1 subscription every two days. A large number of watch events signals high interest from the community of platform users, which signifies the popularity of the project. Summary statistics for the control and instrumental variables used in our study are presented in Online Supplemental Tables 1.6 and 1.7, respectively.

Overall, the monthly changes in the development activities are quite substantial and volatile, allowing us to run the regression analysis to establish the relationship. We further discuss our regression methodology in the next section.

## Empirical Results

We use a panel data fixed effects model to estimate the impact of monthly activity with the source code on the market price dynamics across a panel of cryptocurrencies. We choose a fixed-effects model since it controls for omitted variable bias and time trends (monthly time dummies) due to unobserved heterogeneity. In addition, time-invariant characteristics have been controlled in the fixed effects model by design. Due to the use of the natural logarithms of dependent and independent variables, our fixed effects model framework measures dependency between the change in community development activities and the change in market price, which helps us answer our main research question.

The dataset includes monthly aggregated historical data on market prices of cryptocurrencies as the main dependent variable and the number of open-source activities as independent variables. While the dependent variable is constructed as the data point at the end of the aggregation period (month), the independent variables are aggregated for the period before the price of the focal cryptocurrency is reported. It is important to note that such construction ensures that OSS activities predate the time of the price change. We also construct quarterly aggregates to check for the sensitivity of our results to the time horizon, and the quarterly model yields similar results as the monthly model.

We begin with the estimation of the fixed effects Model (1) to investigate the impact of open-source activities on cryptocurrency prices. However, we recognize that the higher market price can allure more developers and induce higher development effort, and the relationship in focus may suffer from reverse causality. To address these issues and obtain unbiased estimates, we use the instrumental variable (IV) approach in Model (2). We develop our instrumental variable (IV) discussions in the next section.

The fixed effects model looks as follows:

$$ln\_price_{it} = \beta_1 ln\_commits_{it} + \beta_2 ln\_pullreq_{it} + \beta_3 ln\_issues_{it} + \beta_4 ln\_forks_{it} + \beta_4 ln\_watches_{it} + \Gamma Controls_{it} + \Omega TimeDummy_t + u_i + \varepsilon_{it} \quad (1)$$

Here, $t$ is the monthly index, $i$ is the cryptocurrency index, $ln\_price_{it}$ is the logarithm of the price of the cryptocurrency $i$ at month $t$, and the main independent variables are $ln\_commits_{it}$ – the number of code commits, $ln\_pullreq_{it}$ – the number of pull requests, $ln\_issues_{it}$ – the number of issues, $ln\_forks_{it}$ – the number of fork events, and $ln\_watches_{it}$ – the number of watch events. Our $Controls_{it}$ variables are the traded volume $ln\_volume_{it}$ and the maturity of the project by including the variable $Lifetime_{it}$, which is the number of months since the ledger was launched. We add monthly dummy variables $TimeDummy_t$ to account for market trends that are not captured by other variables. Time dummies in Model (1) allow capturing the month-year-specific effects without making a strong assumption about the trend form, that is, linear trends. In addition, time dummies help absorb all other market conditions not captured in the model, including the waves of public searches on the internet and market extremes. Coefficients $\beta_i,\ i = \overline{1,5}$, are estimation coefficients of the ordinary least squares (OLS) regression, $\Gamma, \Omega$ – vectors of coefficients for control and time dummy variables, respectively. The fixed effect, $u_i$, captures the unobserved heterogeneity of cryptocurrency and the co-movement of variables across cryptocurrencies, $\varepsilon_{it}$ – error term.

The fixed effects estimation results are presented in Online Supplemental Table 1.1. Though we explain the results of our first model here, we revise them when conducting the IV analysis. Model coefficients in column (1) are estimated using only the open-source variables of interest. Column (2) shows the estimation results that additionally account for the control variables, and column (3) shows the estimation results that additionally account for control variables and fixed time effects. Following Kutner et al. [81], we gradually add variables into our main model. This allows us to demonstrate stable estimated results and to show that the signs and significance of estimated coefficients for main variables of interest do not change much after new variables are added to the model. The standard errors are clustered at the cryptocurrency level.

We are interested in the coefficients on *ln_commits*, *ln_pullreq*, *ln_issues*, *ln_forks*, and *ln_watches*. Since the specification in the main model uses a log-log model, we can interpret the aggregate coefficients to be the effect on price change from a 1 percent change in the activity in the OSS platform. In Online Supplemental Table 1.1, we observe a negative sign at the *ln_commits* that means as the number of code changes increases, the price of the cryptocurrency will decrease, which is counterintuitive. The negative sign for commits can be explained as the decreasing return on the number of code changes in life-cycle productivity, which negatively affects price change [78]. [110] validates the negative sign of impact for the commits as well.

We also observe that the coefficient on *ln_forks* is significantly positive (column 3, Online Supplemental Table 1.1) that means as the total number of forks on repositories increases by 1 percent, the price of the cryptocurrency increases by 0.041 percent. Our finding implies an increased number of copies of the project in the OSS community is positively associated with the market price of the cryptocurrency. Forks, or copies, are usually created by developers who are willing to make changes to the software code. We treat forks as the popularity metric: The more developers whose skills are highly evaluated on the job market discover the project and interact with it, the more interesting and worth investing time in the project is.

Another metric that is positively related to price is *ln_watches*: as the total number of subscriptions (watchers) in the project's repositories increases by 1 percent, the price of the cryptocurrency increases by 0.088 percent (column 3, Online Supplemental Table 1.1). Watching a repository signals attention to the project [41], and it can be treated as the proxy for the popularity of the project among the general public.

Some of the development activities are inherently related to the software development process, therefore, some concerns regarding their correlation may arise. To check for multicollinearity, we look at the variance inflation factors (VIFs). Most prior studies rely on informal rules of thumb applied to the VIF [6, 26, 76]. Values of variance inflation factors greater than ten are often taken as a signal that the data have collinearity problems [30]. With the results in Online Supplemental Table 1.2, we find that the VIF associated with each variable is below the accepted conventional threshold ($VIF < 10$), indicating that multicollinearity is not an issue in our model and does not significantly impact our findings.

To check for the sensitivity of our results to the time horizon, we construct quarterly aggregates and estimate the model similar to the monthly model. We report model results in Online Supplemental Table 1.3. The sign and significance of estimates are robust. Therefore, the effects of development activities we observe in our main model are stable for monthly and quarterly time horizons.

## Addressing Endogeneity

We employ the IV approach to address the limitation of the OLS model in accounting for possible endogeneity and providing a causal explanation of our relationship. While we look at development activity as the leading indicator of a cryptocurrency's price change, higher (lower) prices of cryptocurrencies could simultaneously enhance (decrease) the appeal to developers. Such simultaneity creates endogeneity and leads to biased coefficient estimates [138]. To mitigate this issue of possible endogeneity in the main model, we implement the two-stage least square (2SLS) approach by identifying IVs. Since there are five endogenous OSS activities, we construct five IVs and get the exactly identified model.

Appropriate IVs must satisfy the following two conditions: (i) they should be correlated with independent focal variables, and (ii) not directly affect the outcome variable, $ln\_price_{it}$, in our context. Following the literature [72], we construct the IVs by aggregating the corresponding activities in other repositories (nonfocal projects) by the same developers who contributed to focal repositories: $ln\_commits\_other_{it}$, $ln\_pullreq\_other_{it}$, $ln\_issues\_other_{it}$, $ln\_forks\_other_{it}$, and $ln\_watches\_other_{it}$. These IVs are constructed based on the following intuition: Typically, OSS platform activity reflects both the developers' preference for the focal project (project-specific factors) and their personal propensity for contributing (individual developer-specific factors). We argue that the instruments that we pick satisfy the two conditions for an appropriate IV. First, the number of activities in other projects is generated by the same set of developers and is intrinsically related to the volume of activity in the focal project; therefore, we argue that condition (i) is satisfied. Second, we believe that activities in other GitHub projects are generally not related to the focal cryptocurrency price dynamics. For example, if the same developers may have contributed to Linux projects and cryptocurrency projects, only the former ones are accounted in the IV. Since the overall market condition of cryptocurrencies should not be attached to the development of Linux projects, we can conclude that our IVs are possibly exogenous and are not driven by the price of the focal project.

Since the unit of our observation is cryptocurrency-month, the contributions of all developers in a particular project have been aggregated (summed) over a month. On average, 46 percent of developers who contribute to the focal cryptocurrency repositories also contribute to other nonfocal repositories. Due to the diversity of developers (narrowly focused on one cryptocurrency and broadly focused on OSS development) in each project, we do have valid (non-null) values of IVs for our study. The summary statistics table for IVs is in Online Supplemental Table 1.7.

We apply the generalized method of moments (GMM) to estimate the FE IV Model (2):

$$\begin{aligned} ln\_price_{it} = \beta_1^* ln\_commits_{it}^* + \beta_2^* ln\_pullreq_{it}^* + \beta_3^* ln\_issues_{it}^* + \beta_4^* ln\_forks_{it}^* \\ + \beta_5^* ln\_watches_{it}^* + \Gamma Controls_{it} + \Omega TimeDummy_t + u_i + \varepsilon_{it} \end{aligned} \tag{2}$$

where $\beta_i^*,\ i = \overline{1,5}$, - estimation coefficients of the second stage regression, $\Gamma, \Omega$ – vectors of coefficients for control and time dummy variables of the second stage regression, respectively.

Here, first-stage regressions are of the form:

$$\begin{aligned} & ln\_activity_{it}^* = \alpha^{IV} + \alpha_1 ln\_commits\_other_{it} + \alpha_2 ln\_pullreq\_other_{it} + \\ & \alpha_3 ln\_issues\_other_{it} + \alpha_4 ln\_forks\_other_{it} + \alpha_5 ln\_watches\_other_{it} + \\ & \Gamma^{IV} Controls_{it} + \Omega^{IV} TimeDummy_t + \xi_{it} \end{aligned} \tag{3}$$

where $ln\_activity_{it}^* = \{ln\_commits_{it}^*, ln\_pullreq_{it}^*, ln\_issues_{it}^*, ln\_forks_{it}^*, ln\_watches_{it}^*\}$, $\alpha^{IV}$ – estimated OLS constant, $\alpha_i,\ i = \overline{1,5}$, - OLS estimation coefficients for instrumental variables of the first stage regression, $\Gamma^{IV},\ \Omega^{IV}$ – vectors of OLS coefficients for control and time dummy variables, respectively, $\xi_{it}$ – error term.

We are interested in the coefficients on *ln_commits*, *ln_pullreq*, *ln_issues*, *ln_forks*, and *ln_watches*. Since the specification in the main model uses a log-log model, we interpret the aggregate coefficients as the effect on price change from a 1 percent change in the activity in the OSS platform. We cluster robust standard errors at the cryptocurrency level. Due to the appropriateness of the empirical method and the significance of the results, we rely on the IV model estimates to explain the statistical and economic effects of the OSS activities on the cryptocurrency market price. We present the results of the estimation in Table 3.

An interesting difference between FE and FE IV is that significance of commits and pull requests has flipped: While FE commits were significantly negative predictors of the price, in FE IV, this function is transferred to pull requests (Table 3). Controlling for endogeneity, we now observe that the code suggestions by outside developers form a negative sentiment rather than code changes made by core developers. The reverse dependency between suggested code contributions (pull requests) and the token price is counterintuitive and requires further analysis. On the one hand, as developers suggest more code, it signifies higher community involvement and willingness to contribute. On the other hand, the increased number of submissions creates delays in the software code deployment due to the review process bottleneck. A large number of pull requests may also be a distinctive

**Table 3.** IV parameter estimation.

| Variables | (1) OLS model | (2) IV model |
|---|---|---|
| ln_commits | -0.0170* | 0.0182 |
| | (-1.872) | (0.594) |
| ln_pullreq | 0.00406 | -0.110*** |
| | (0.310) | (-2.589) |
| ln_issues | 0.0190 | 0.0768** |
| | (1.567) | (2.364) |
| ln_forks | 0.0408*** | 0.0868*** |
| | (3.354) | (4.646) |
| ln_watches | 0.0876*** | 0.0396*** |
| | (6.682) | (2.702) |
| Constant | -0.0852 | -0.0831 |
| | (-1.085) | (-1.066) |
| Cryptocurrency FE | Yes | Yes |
| Control variables | Yes | Yes |
| Monthly FE | Yes | Yes |
| Observations | 15,716 | 15,716 |
| Number of cryptocurrencies | 559 | 559 |
| Within R-squared | 0.294 | 0.270 |
| Between R-squared | 0.088 | 0.073 |
| Overall R-squared | 0.094 | 0.075 |

*Notes*: Robust z-statistics in parentheses. Abbreviations: OLS, ordinary least squares; IV, instrumental variable; FE, fixed effects. ***$p < 0.01$. **$p < 0.05$. *$p < 0.1$.

feature of the development culture of the contributors and project, i.e., more experienced and skilled developers would contribute fewer times but include more changes with each pull request. In contrast, the less experienced developers would oversee important changes at first and contribute in multiple batches. Overall, as the total number of pull requests increases by 1 percent, the cryptocurrency price decreases by 0.11 percent. Given the one standard deviation increase in monthly pull requests (62.5), the price of the cryptocurrency will decrease by 0.45 percent monthly on average (-0.011 percent* ln(63.5)) or around 5 percent annually. Preventing the 5 percent loss of the individual portfolio can yield quite substantial economic benefits depending on the size of the investment.

We also observe that while remaining positive, the coefficient for the number of issues becomes significant in the FE IV model. The coefficient on *ln_issues* is significant and positive (column 2, Table 3), which means as the total number of issues increases by 1 percent, the price of the cryptocurrency increases by 0.076 percent. This result is noteworthy, as the larger number of issues signifies a larger number of recommendations for improvement that is the corollary of the higher cryptocurrency adoption. Given the one standard deviation increase in monthly issues (119.1), the price of the cryptocurrency will increase by 0.36 percent monthly on average (ln(119.1)* 0.076 percent) or around 4.4 percent annually. Assuming the investor holds the portfolio throughout the year, it would earn at least a 4.4 percent return per annum.

Both popularity metrics, the number of forks and watches, remain significant and positive (column 2, Table 3), although they change their magnitude in the FE IV model.

The magnitude of the *ln_forks* coefficient in the IV model is significantly larger than in the main model and means as the total number of forks increases by 1 percent, the price of the cryptocurrency increases by 0.086 percent. Our finding implies an increased number of copies of the project in the OSS community is positively associated with the cryptocurrency's market price. Forks, or copies, are usually created by developers interested in the project and willing to make changes to the software code. Given the one standard deviation increase in monthly forks (61.5), the price of the cryptocurrency will increase by 0.35 percent monthly on average (0.086 percent* ln(61.5)) or around 4.3 percent annually.

Based on the IV model estimates, we calculate the relative additive effect of quality versus popularity signals. Given the one standard deviation increase in monthly pull requests (62.54) and issues (119.10), the total effect of a quality improvement effort will be -0.09 percent monthly, or -1.1 percent annually on average. Given the one standard deviation increase in monthly forks (61.48) and watches (168.60), the total effect from a popularity increase will be 0.56 percent monthly, or 6.7 percent annually on average. The ratio between the two indicators is 6.38, which signifies that popularity signals are 6 times more effective in signaling the superior underlying quality of the software project rather than the power of quality signals to indicate existing problems with the software code (e.g., existing bugs or technical debt). Therefore, while it is easier to bookmark a good project (fork, watch), development takes longer time, and software quality indicators such as pull requests may be lagging in time. Investors should remember that popularity activities (fork, watch) are fairly easy to perform, and they should recalibrate their investment strategies based on both, dynamics of quality and popularity signals and their relative ratio.

Another metric with positive sign and significance is *ln_watches*: as the total number of subscriptions (watchers) in the project's repositories increases by 1 percent, the price of the cryptocurrency increases by 0.039 percent (column 2, Table 3). This effect is smaller than in the main model (0.087 percent), however, bears a significant economic effect. Given the one standard deviation increase in monthly watches (168.6), the price of the cryptocurrency will increase by 0.19 percent monthly on average (0.039 percent* ln(168.6)) or around 2.4 percent annually.

Moreover, as we can see from Online Supplemental Table 1.4, platform activities performed in other projects are a good predictor of the activity in the focal repository, and all F-statistics in first stage regressions are reasonably high to confirm a strong relationship between the endogenous variables and instrumental variables. Online Supplemental Table 1.4 presents Stage I regression coefficients and F-statistics from all models. We also test the IV model for weak instruments using the Kleibergen-Paap rk Wald F statistic [11]. The conventional rule for Kleibergen-Paap statistic is to be at least 10 for weak identification not to be considered a problem [128]. The identification test reports large values of Kleibergen-Paap Wald statistic (29.597), which signifies that the model is well-identified and selected instruments are correlated with endogenous variables.

Our results confirm a connection between open-source activity and the market price of the crypto asset. The market price reflects the expectations of the future valuation of the cryptocurrency. Commonly, for traditional publicly traded companies, the price increase reflects the increase in stock demand induced by the higher expectation of the company's quality, which is measured by the quality of its operations. In the cryptocurrency case, the quality of operations is defined by the underlying algorithm and software code that runs it. We provide a deeper analysis of the cryptographic algorithm and its moderating effect in the Online Supplemental Appendix 3. For the cryptocurrency project managers, our results can be used as evidence of price feedback in response to OSS development contributions. Our result can motivate the development teams to invest more effort in project development.

## Robustness Checks

### Alternative Specifications

#### Random Effects Model

We estimated our model with several alternative specifications. First, we estimate a random-effects model via maximum likelihood (RE-ML) and conduct a Hausman test for model specification [65]. Hausman-like overidentification test identifies if the regressors are uncorrelated with the group-specific error, and the random-effects model should be chosen instead [7, 119]. The estimated Sargan-Hansen statistic is 3,189, and the p-value = 0.000. Therefore, the hypothesis about the correlation between regressors and group-specific error is rejected, and the fixed effects model should be chosen for further analysis.

#### Panel Vector Autoregression

Next, we employ the panel vector autoregression model (PVAR) model to elaborate on the signaling mechanism of the established OSS development impact. While the OLS Model (1) and the IV Model (2) allow us to capture short-term dependencies between development effort and the market response, the long-term interdependencies between independent and dependent variables remain convoluted. As a type of simultaneous equations model, the PVAR model allows disentangling the impact of independent and dependent variables onto each other and tests the impact of exogenous shocks on the system in a long-term perspective. Hence, giving researchers the opportunity to trace the whole path of impacts identified in the causal models. The PVAR model results are complementary to the IV model in the sense that both models are different in their design and not conflicting with each other.

To further examine underlying mechanisms between open-source activities and cryptocurrency market prices, we estimate the PVAR model. PVAR helps us understand the complexity of the dynamics between endogenous variables and capture the feedback loops. In addition, the PVAR model ensures the robustness to issues of nonstationarity, spurious causality, endogeneity, serial correlation, and reverse causality [3] that benefits the study. The PVAR model is presented in Eq. (4).

$$\begin{pmatrix} ln\_price_{i,t} \\ ln\_commits_{i,t} \\ ln\_pullreq_{i,t} \\ ln\_issues_{i,t} \\ ln\_forks_{i,t} \\ ln\_watches_{i,t} \end{pmatrix} = \sum_{p=1}^{s} A_p \begin{pmatrix} ln\_price_{i,t-p} \\ ln\_commits_{i,t-p} \\ ln\_pullreq_{i,t-p} \\ ln\_issues_{i,t-p} \\ ln\_forks_{i,t-p} \\ ln\_watches_{i,t-p} \end{pmatrix} + B(Controls_{it}) + u_i + \varepsilon_{it} \tag{4}$$

The PVAR model treats all variables as endogenous, and a vector of time series variables is regressed on lagged vectors of these variables. Vector $Y_{i,t} = (ln\_price_{i,t},\ ln\_commits_{i,t},\ ln\_pullreq_{i,t},\ ln\_issues_{i,t}, ln\_forks_{i,t},\ \ ln\_watches_{i,t})$ is a column-vector of dependent variables for each cryptocurrency at time $t$. $Y_{i,t}$ contains all variables that are likely to be endogenous. $A_p$ is a matrix of coefficients for $p$-period lagged dependent variables, and $s$ is the maximum number of lags. We also include other control variables without the lag and estimate their parameters $B$.

In our specification, we assume that current shocks to the open-source activity affect the change in the market price of the digital token, while market price change has an effect on the marginal contributions to the open-source repositories with a lag. We believe this assumption is plausible for two reasons. First, the development process on the open-source platform starts even before the token is launched on the market or before the demand and price is formed. Second, software development requires some time to create visible and documented feedback to the price movement. Further, the project development process follows the pre-defined strategy and is not necessarily reactive to short-term market movements.

We have a panel dataset with project-specific fixed and time effects, so we use the system-GMM to obtain consistent and efficient estimates [20]. The method uses the lagged differences of the dependent variable and the lagged levels of the dependent variable[13] as regressors in equations in levels. The first-differencing procedure was applied to remove the panel fixed effects, and system-GMM was applied to get the unbiased coefficients [8]. Time effects were eliminated by subtracting the means of each variable calculated for each cryptocurrency-month. In addition, we check for the stability of PVAR using the unit root test.[14] The stability conditions are satisfied since all the eigenvalues of the companion matrix lie in the unit circle (Online Supplemental Figure 1.13).

Table 4 shows the dynamics among OSS contribution and market price. We first examine the results of the regressions with closing price as the dependent variable to see how the OSS development efforts would influence the market dynamics. We observe the positive autocorrelation in price up to the third lag. We also observe interesting patterns with respect to the delayed effects of OSS effort: The third lag of pull requests has a negative and 90 percent-significant effect on the cryptocurrency monthly price. Interestingly, the issues additionally have a positive and significant effect on the market price with one lag delay, and the forks have a positive and significant effect with two and three-lag delays. The significance of these variables is consistent with the IV model results in Table 3. Further investigation of the model results allows us to shed light on the underlying mechanisms of impact.

Based on the IV model (Table 3) and PVAR model estimates (Table 4), we suggest there is a statistically significant lagged relationship between OSS signals of popularity and quality. For example, the number of pull requests in period t increases if the number of new subscriptions (watches) grows in (t-1) and (t-2) (column 3, Table 4). This signifies that popularity growth, which is positively associated with the market price, will inevitably convert to the activity of code improvements (pull requests) and the market sentiment may change due to the negative effect of pull requests growth on market price (Table 3). Strategic investors should closely monitor both categories of metrics to calibrate their strategies when the magnitudes of monthly OSS activities change substantially. Thus, the

**Table 4.** PVAR(3) Model estimates.

| Variables | (1) ln_price$_{i,t}$ | (2) ln_commits$_{i,t}$ | (3) ln_pullreq$_{i,t}$ | (4) ln_issues$_{i,t}$ | (5) ln_forks$_{i,t}$ | (6) ln_watches$_{i,t}$ |
|---|---|---|---|---|---|---|
| ln_price$_{i,t-1}$ | 0.116*** | 0.924*** | 0.419*** | 0.454*** | 0.346*** | 0.698*** |
| | (4.778) | (10.64) | (8.736) | (9.382) | (6.851) | (13.09) |
| ln_price$_{i,t-2}$ | 0.178*** | -0.280*** | -0.0417 | 0.0408 | 0.245*** | 0.268*** |
| | (7.952) | (-3.921) | (-1.025) | (0.978) | (6.086) | (6.161) |
| ln_price$_{i,t-3}$ | 0.185*** | 0.160** | -0.0636 | 0.0115 | 0.131*** | 0.281*** |
| | (9.648) | (2.417) | (-1.626) | (0.318) | (3.868) | (7.329) |
| ln_commits$_{i,t-1}$ | 6.74E-05 | -0.357*** | 0.0470*** | 0.0321*** | 0.0380*** | 0.0489*** |
| | (0.0255) | (-20.66) | (4.828) | (3.578) | (3.929) | (4.897) |
| ln_commits$_{i,t-2}$ | -0.00427 | -0.261*** | 0.0267*** | 0.0297*** | 0.0237** | 0.0339*** |
| | (-1.642) | (-15.08) | (2.691) | (3.096) | (2.346) | (3.246) |
| ln_commits$_{i,t-3}$ | 0.00261 | -0.142*** | 0.0219** | 0.0219** | 0.0173** | 0.0283*** |
| | (1.039) | (-9.490) | (2.386) | (2.534) | (1.965) | (3.076) |
| ln_pullreq$_{i,t-1}$ | -0.00204 | -0.0292 | -0.459*** | -0.0300* | -0.0126 | -0.0398** |
| | (-0.525) | (-1.164) | (-23.96) | (-1.922) | (-0.782) | (-2.422) |
| ln_pullreq$_{i,t-2}$ | 0.00412 | 0.0133 | -0.273*** | -0.0157 | -0.0121 | -0.0520*** |
| | (0.933) | (0.525) | (-14.25) | (-0.907) | (-0.733) | (-3.164) |
| ln_pullreq$_{i,t-3}$ | -0.00728* | -0.0142 | -0.159*** | -0.0383** | -0.0522*** | -0.0511*** |
| | (-1.923) | (-0.629) | (-9.048) | (-2.571) | (-3.430) | (-3.337) |
| ln_issues$_{i,t-1}$ | 0.00880** | 0.0876*** | 0.0845*** | -0.424*** | 0.0662*** | 0.0600*** |
| | (2.345) | (4.075) | (5.172) | (-22.44) | (4.703) | (3.861) |
| ln_issues$_{i,t-2}$ | -0.000265 | 0.0808*** | 0.0681*** | -0.244*** | 0.0474*** | 0.0489*** |
| | (-0.0677) | (3.543) | (4.01) | (-12.60) | (3.204) | (3.076) |
| ln_issues$_{i,t-3}$ | -0.00528 | 0.0259 | 0.0186 | -0.122*** | 0.0434*** | 0.0326** |
| | (-1.353) | (1.19) | (1.255) | (-7.741) | (3.275) | (2.29) |
| ln_forks$_{i,t-1}$ | 0.00421 | -0.0127 | -0.00472 | 0.0266** | -0.607*** | 0.0801*** |
| | (1.235) | (-0.652) | (-0.333) | (1.973) | (-36.96) | (5.331) |
| ln_forks$_{i,t-2}$ | 0.00882** | -0.018 | -0.0279* | 0.0185 | -0.371*** | 0.0572*** |
| | (2.34) | (-0.814) | (-1.765) | (1.281) | (-23.01) | (3.63) |
| ln_forks$_{i,t-3}$ | 0.00966*** | -0.0254 | -0.0361*** | 0.0167 | -0.178*** | 0.0576*** |
| | (2.7) | (-1.272) | (-2.614) | (1.346) | (-13.05) | (4.268) |
| ln_watches$_{i,t-1}$ | 0.00126 | 0.0856*** | 0.0385*** | 0.0304** | 0.0850*** | -0.594*** |
| | (0.389) | (4.357) | (2.837) | (2.313) | (5.849) | (-36.53) |
| ln_watches$_{i,t-2}$ | -0.0017 | 0.0994*** | 0.0395*** | 0.0432*** | 0.0607*** | -0.372*** |
| | (-0.452) | (4.657) | (2.809) | (3.119) | (4.035) | (-22.34) |
| ln_watches$_{i,t-3}$ | -0.00112 | 0.0324* | 0.0234* | 0.0466*** | 0.0643*** | -0.151*** |
| | (-0.375) | (1.808) | (1.912) | (4.035) | (5.24) | (-11.12) |
| Control variables | YES | YES | YES | YES | YES | YES |
| Time dummies | YES | YES | YES | YES | YES | YES |
| Observations | 9,924 | 9,924 | 9,924 | 9,924 | 9,924 | 9,924 |

*Notes*: Robust t-statistics in parentheses.
***p < 0.01. **p < 0.05. *p < 0.1.

intertemporal dependency between the activities with positive and negative signaling properties reflects the dynamic nature of the software quality [105]. From the IV model analysis, we know that on average the signaling power ratio between the popularity and quality signals is 6.38. In a constantly evolving environment like ours, it is the ratio of signals that creates a separating equilibrium at any given point in time.

The estimation results show that the coefficients of some market price lags on OSS activities are also positive and significant. Generally, this means that the larger the price increase in the preceding months, the more numerous the OSS activity in the current month. Together with our IV estimation, the PVAR model results illustrate that there is a feedback relationship between OSS activities and market prices: A higher level of OSS activities tends to drive up market prices, and vice versa. Recall that to estimate the causal impact of OSS activity on market prices, we use an IV estimation strategy to introduce an exogenous shock that would break this feedback loop. The combined results from our IV and PVAR models demonstrate a positive impact of OSS activities on market prices, even after accounting for the fact that higher market prices can lead to increased OSS activities.

Overall, our findings suggest a strong and significant link between OSS activities and market prices. For example, there is immediate positive feedback between market price increase in a month (t-1) and pull requests and issues in month t (columns 3 and 4, respectively, Table 4). The significance of the price coefficients with lags (t-2) and (t-3) signifies that there is the positive and long-term effect of price increase on forks and watches (columns 5 and 6, respectively, Table 4). One of the reasons for such positive relationships is that price increases usually attract public attention to the cryptocurrency and lead to higher engagement with their web profiles including GitHub account. Therefore, voluntary OSS activities increase. Conversely, activities that require extended permissions to make changes to the project, such as commits, are not straightforward to contribute to due to the peer review process. The number of new commits may fluctuate depending on the complexity of the pull requests awaiting review. Hence, the changing sign of the price lags in column 2 of Table 4 might reflect the congestion in the review system. Therefore, PVAR results allow us to conclude that OSS activities not only form short-term sentiment but significantly reflect long-term tendencies in cryptocurrency quality formation. Incremental development effort creates a strong perception of software quality as one of the fundamentals for digital assets.

We also conducted an impulse-response function (IRF) analysis to examine the long-term effects. IRFs estimate the response of one endogenous variable to the one-unit shock in the other variable [130]. The ordering of the variables is aligned with the construction of our variables and looks as follows: commits, pull requests, issues, forks, watches, and market price. This ordering means that the OSS activities affect the market price contemporaneously (same time index) and with a lag, but the market price affects other variables only with a lag.[15] Figure 3 illustrates the results of IRFs for our main effect model.

We are particularly interested in the reaction of price to the shocks in the OSS activities (Figure 3). Surprisingly, the IRF analysis suggests that the reaction to the shock in the number of commits is instant and does not have a lagged effect. The pull requests create a negative response in the market price three months after the shock. The delayed response can be explained by the delay in the review process of the pull requests: It requires time to review changes, make the acceptance decision, and turn suggestions into code innovation. Code innovations lead to higher security and better functionality of the cryptocurrency software; therefore, it is important to carefully review adjustments to the code.

The number of issues creates an immediate negative response in price and a positive response after one month. Issues are instantly associated with bugs, therefore, the negative effect on price. After an issue is created, it captures the attention of the developers, and the expressed concern or recommendations are resolved in code, which enhances the quality of

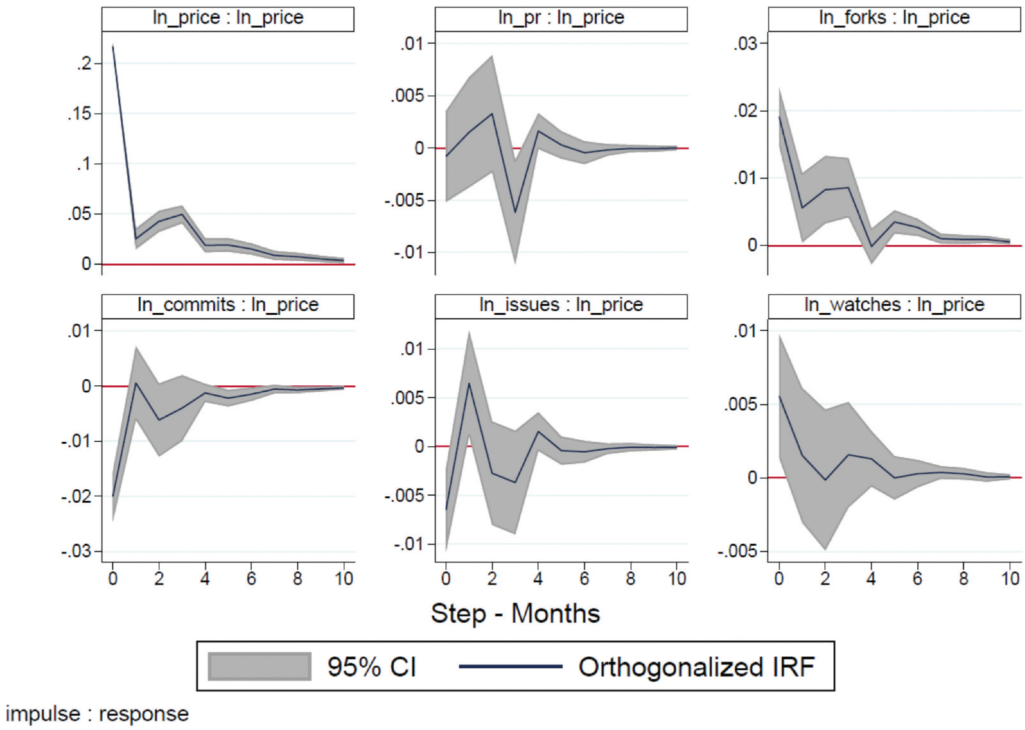


**Figure 3.** Impulse response function result.

the cryptocurrency software and creates a positive image. The delay is related to the development practices, especially in the OSS environment: First, after the issues are posted on the project's page, they should be assigned to responsible developers who will address the issue. Such assignments will not happen immediately due to the ongoing work; however, the issues will be eventually addressed by the responsible developers, and appropriate improvements to the software will be made.

The number of forks creates an immediate positive response in price, and the positive effect lasts for about five months after the shock. Second, forks have a longer lag due to a longer processing cycle: first, the change to the copied code should be performed; after which it will be pushed in the form of the pull request to the main repository, and finally, the pull request will be peer-reviewed by the responsible developers for the changes to be approved or denied. Finally, the number of watches has a significant positive effect on market price within the same month only. Watches reflect the dynamics of attention to the whole project in a particular month, and it is not obvious if the audience that attended a particular project in the last month is also paying attention in the current months. Therefore, we observe the short-lasting effect of our second popularity variable. Our results are consistent with the findings in the IV model (Table 3).

Our result corresponds to the common sentiment from practitioners (e.g., Mihov [102]). The insights from IRFs provide important implications for investors, cryptocurrency platforms, and cryptocurrency project managers. For example, when inspecting the OSS development of the cryptocurrency project, investors should not only pay attention to the most recent number of contributions but also look at the historical dynamics of social activities (pull requests, issues, and forks) that will turn into quality improvements in the future. Second, in their attempt to inform potential investors, cryptocurrency platforms should provide historical values of the main variables and clearly explain the dynamic effects to avoid misrepresentation [111]. Finally, because development activities will take around three months to be reflected in the price, managers of the OSS cryptocurrency projects should maintain the contribution processes on high levels and timely assign social suggestions, issues and pull requests to the responsible developers for review. Losing reputation due to the drops in development activity can lead to negative effects much larger in scale than previously acquired gains in reputation [32].

## Discussion

In this study, we investigate the impact of open-source code development activity on the price of cryptocurrencies. Despite its technological nature, OSS development is a collective and public process [67], and the community dynamic on OSS platforms resembles the dynamics on other social media platforms, such as Twitter or Reddit. Users on all three platforms tend to share information, engage in social activities such as following, and develop discussion threads [105, 124, 131]. The features of the OSS platform allow registered developers to interact with other developers by reviewing their pull requests, express their concerns regarding the software features through issues, reuse (cite) pieces of code in their own development (forks), and favor and follow the project through subscription (watches).

On the other hand, prior research points to significant differences in the information diffusion properties of the three networks. [124] find that diffusion in GitHub is stronger due to the denser network structure that helps to spread the information faster. In addition, the interpretation of the public activities on the OSS platform is different from the activity on other public platforms: Most of the public platforms are designed as forums, i.e., where participants can engage in verbal discussion, whereas on the OSS platforms developers engage in the practical activity that enhances the cryptocurrency software [61]. In other words, the OSS platform users (developers) focus more on product enhancement, while forum participants (social media users) focus more on information diffusion and social contagion [5, 94].

Our results have the potential to be generalized for other types of innovative products and services that require effort consistency and build their quality over time. By focusing on the unique context of cryptocurrencies, our study fills the gap in the Information Systems literature by providing an understanding of various types of interactions with the source code and their effects on the market value of the business.

### *Implications for Research*

Our study complements the IS literature by pointing to the OSS development strategies to reduce information asymmetry. In established markets, detailed analysis of financial, economic, and market data allows the reduction of information asymmetry between buyers and sellers of financial assets. Whereas in emerging markets, the scarcity of historical financial data creates a much larger discrepancy between sellers and not-so-well-informed buyers. Therefore, searching for new indicators pertinent to the digital product is crucial for practitioners and academics. In the case of cryptocurrencies, signaling about the product through open-source software code is a common tool. To the best of our knowledge, this study is one of the first to treat activity in the developers' community as a viable investment signal.

We also complement prior research on market signals [21] by illustrating the effects of OSS development efforts on the market evaluation of the cryptocurrency. We establish empirical evidence of the significance of OSS activities and explain the causal mechanism of their relationship to the market price by relating the OSS effort information to the perceived software quality and digital asset demand. The "rule of code" in the digital age [93] urges modern investors to understand how the software functions and how it is being developed. The idea of digital literacy is also supported by Leonardi and Neeley [85], which embraces the ability to recognize the software nature of cryptocurrencies and the software development process in general.

We extend our contribution to the OSS literature by providing meaningful explanations of the existent signaling mechanisms in the OSS community. Our study connects with Aboody and Lev [2], who show that software development itself can be considered an asset and is linked to market capitalization and earnings. Aboody and Lev [2] show that software development cost is positively related to the future company performance, in both cases when the firm chooses to report it as the software development expenses ("expensers") or software asset value ("capitalizers"). The higher software development cost signals a higher number of successful projects (that passed the feasibility test) and hence investors would inflate their expectations about the company. In a similar vein, our research illustrates that projects with higher development efforts perform better in terms of market performance due to the larger community effort and better software product quality. The contribution of our study is that we extend the mechanism of the software development cost impact on investor decision-making in circumstances when the reporting standards are not set yet and conventional reported indicators are not available, that is, the cryptocurrency market. Despite the absence of unified reporting standards in the crypto industry, investors seek signals to inform their decision-making, and the OSS software development effort serves as a unified metric of software development quality across different projects. In addition, the unique context of our study allows us to illustrate this relationship in an open-source development context, where the drivers of community contribution differ from the proprietary software development.

Finally, our study extends the emerging blockchain literature by investigating open-source contributions as a factor of price determination in line with previous findings on the signaling power of patents and human capital for technology startup potential [12].

### *Implications for Practitioners*

The ability to obtain insights from observable characteristics of the development community is crucial for the crypto and tech industry [113]. We provide practical insights for a nontechnical audience who seek a better understanding of the signals from the software development community. First, our study provides evidence for investors about the existing relationship between the open-source software community and the market price of the cryptocurrency. We identify that commits and pull requests are significant predictors of the price decrease and issues, forks, and watches are signals for the price increase. The identified dependencies may be included in the portfolio-building strategies by investors alongside other market signals. Moreover, our analysis can be used by the crypto analysts included in their reports [46].

Second, cryptocurrency project managers can use the outcomes of our research as a motivational tool for their developers. The feedback between the developer's contributions and the market price could be demonstrated to contributors as a reward appealing to their intrinsic motives. In addition, in case developers are token holders, good maintenance of the project and high OSS activity will be rewarded by the growth of their own portfolio. New entrants and incumbents can adjust their focus to deliver higher quality products and perceptions of the market value through more thorough attention to the disclosed development process.

Based on our results, the market performance of cryptocurrencies attracts contributors to their OSS projects and generates traffic on the OSS platform. Therefore, platforms like GitHub should focus on the platform design and policies that maintain high-quality projects and retain skilled labor on the platform, similar to practices on other major platforms [80, 111]. The traffic of the OSS platform is made up of all types of open-source activity. Platforms with higher traffic appeal to developers more so that they have more collaboration opportunities. Increased skilled labor supply attracts more projects which try to utilize available resources.

Finally, understanding cryptocurrency quality formation is vital for developing new regulation strategies by policymakers. A combination of the previous research on cryptocurrencies and the discovery of the underlying factors of price formation may guide policymakers to ensure high industry standards and investor protection. Explanation of the quality and the underlying value of the cryptocurrency is an important factor in creating regulatory policies in this sector. We suggest that the formulation of the framework for information disclosure in the emerging sector will provide a more congruent signal and incorporate information from the market more fully for investment decisions.

## Limitations and Future Research

In this study, we investigate the effects of development activities on the market price of cryptocurrency. Our analysis focuses on the sample of cryptocurrencies that have their source code publicly available on the OSS platform. Nevertheless, we recognize the limitation of our study due to the unobservability of the development activity in the projects which code we cannot observe. We suggest that this concern is alleviated by the observable tendency of the majority of cryptocurrencies (66 percent) to be open-sourced. Moreover, open-sourced cryptocurrencies draw more attention from investors and, hence, require first-hand analysis of their activity (Online Supplemental Table 1.8).

There are several possible extensions to our research. First, our study reveals the effects of open-source contributions on the cryptocurrency's market price. Future studies using proprietary data could focus on how software improvement impacts the market price of the software good, given the growing technical awareness of users. Second, we use the number of developer actions as our primary focus variables. Similar to the prior research [31], we realize there might be a quantity-quality tradeoff, and commits may not be homogeneous in quality and other activity indicators. We suggest that a deeper analysis of the quality of developers' actions could be done. Third, in our study, we do not account for the skill levels of contributors. The skill set and experience of developers who maintain the project could significantly impact forking activity. It would be interesting to account for their characteristics as well in the future. Additional studies can be done to estimate the effects of code reuse and technological innovation added in each subsequent copy on the market success of the cryptocurrency (Online Supplemental Appendix 2).

## Conclusion

Cryptocurrencies emerged as software projects and gained popularity as an investment tool. Their dual nature motivates us to examine the relationship between the software quality and market dynamics of cryptocurrencies. Based on the prior signaling and OSS literature, we recognize that the open-source software development effort has to signal properties of future market movements. Using data from the largest OSS platform GitHub and data from one of the most popular cryptocurrency price trackers CoinMarketCap, we establish the connection between open-source activities and the price movement of cryptocurrency. In particular, as project popularity (forks and watches) and users' feedback (issues) increase, the market price increases by 4.3 percent, 2.4 percent, and 4.4 percent per annum respectively. On the contrary, the number of code corrections (pull requests) is negatively related to prices leading to a 5 percent annual price decrease. The underlying mechanism is related to the OSS signals' property to create a separating equilibrium between the cryptocurrencies of differentiated quality. Based on the prior OSS literature, we theorize that the cryptocurrency projects with better product market offerings attract more development talent that reflects the dynamics of the OSS metrics. Our results suggest that OSS contributions create a perfect selection mechanism, where higher quality projects receive more developer attention and user feedback, whereas lower quality projects do not, thus creating the separating equilibrium. Moreover, our setting allows us to illustrate the dynamic and evolving nature of the software quality. In the OSS community, development activities aimed at improving quality (pull requests) depend on the values of subscriptions (watches) and forks in prior periods up to lag of two periods, (t-2). We conjecture that there is a causal relationship between quality and popularity signals and that the market weighs in both types of signals. Based on the empirical evidence, we also compute the signaling power ratio between the popularity and quality signals and identify that the positive effect of the popularity signals is six times stronger than the negative effect of the quality signals on price.

Our study contributes to the signaling literature by revealing the relative power of the positive popularity signals and negative quality signals in the context of the OSS activities in cryptocurrency software projects. The prior OSS literature has suggested the importance of observing the social contribution activities in the OSS projects to estimate their quality [41]. We contribute to this strand of literature by extending the notion of the OSS projects to cryptocurrency projects with explicit monetary evaluations. Hence, our study uses a unique setting where the OSS project has market valuation and estimates the effect of OSS development and social activities on its market value. Moreover, we contribute to the rapidly growing literature on cryptocurrencies and suggest that OSS activities are one of the fundamental factors of their quality and should be closely evaluated and accounted for in the financial evaluation models of cryptocurrencies.

Our findings allow us to formulate valuable recommendations for investors, cryptocurrency project managers, and policymakers. Investors can use our empirical results to factor in the OSS activities in their investment strategies. Cryptocurrency project managers can use our insights about the intertemporal dependencies of the OSS activities and reach out to developers to lure their attention and skillset to provide feedback to their projects. OSS platforms, like GitHub, can use our results as evidence of the importance of transparency on the platform and features that allow us to verify the contributions to different OSS projects. Finally, policymakers should use this evidence to formulate information disclosure policies regarding technological projects such as cryptocurrencies, algorithms, and other technological ventures.

## Notes

1. Total amount estimated by authors based on [97].
2. #51 most visited website in Finance/Other Finance category in the US (https://www.similarweb.com/, accessed June 14, 2023).
3. #15 most visited website in Finance/ Investing category in the US (https://www.similarweb.com/, accessed June 14, 2023).
4. Calculated by authors using CoinMarketCap data on August 31, 2018. Based on the data sample, 1,274 projects had their GitHub link posted on their page out of 1,934 cryptocurrencies listed.
5. https://docs.github.com/en/get-started/quickstart/set-up-git (accessed June 14, 2023).
6. Open-source Unix-like operating system (https://en.wikipedia.org/wiki/Linux, accessed June 14, 2023).
7. See www.coinmarketcap.com (accessed June 14, 2023).
8. See www.github.com (accessed June 14, 2023).
9. Note that we refer to forking as the activity at GitHub. Social coding involves contributing to someone else's project by creating a personal copy of someone's code, and a fork is used to make changes without modifying the original code. Hence, forking on GitHub is merely a convenient instrument for developers to coordinate their work. Nevertheless, sometimes forked cryptocurrency projects can be turned into new cryptocurrencies (hard forks). For this to happen, software changes made via forking code from GitHub repositories require consensus from the blockchain network. Therefore, we distinguish between GitHub workflow forks and cryptocurrency hard forks. While forking in the OSS platform is a usual component of the workflow, forking GitHub does not necessarily entail forking crypto.
10. Google's serverless, highly scalable enterprise data warehouse. https://cloud.google.com/bigquery/ (accessed June 14, 2023)
11. Figure 2 is constructed by the authors using the GitHub Archive data for the respective period.
12. Daily closing price is reported as volume-weighted average of market pair prices for the cryptocurrency, denominated in USD, at 11:59 pm UTC of the corresponding date.
13. Typically, the maximum lag length is selected based on the comparison of statistics that minimizes the trade-off between the model fit and the number of parameters from models with different lag lengths. We follow the Andrews and Lu [4] approach and select the optimal lag for the model selection based on *J*-criterion (similar to information criteria like BIC and HQIC). The third-order panel VAR was chosen. Selection procedure results are reported in Online Supplemental Table 1.5.
14. The stability condition implies that the panel VAR is invertible and has an infinite-order vector moving-average representation, providing a known interpretation of estimated impulse-response functions [3, 96]. We use the *pvarstable* function to find the characteristic roots and check if they are equal to or greater than 1.

15. We estimate the IRFs up to a 10-month lag. The impulse responses are derived from a Cholesky decomposition to orthogonalize the shocks, i.e., make them mutually uncorrelated.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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